# Objective 2 - Uttarakhand: summary

*Simulation, algorithms, reasons, speed-up and results in brief. Full version: Objective2_Uttarakhand_Algorithms_Explained.docx. 9 November 2026.*

**In one paragraph.** For each of Uttarakhand's 4 climate regimes from Objective 1, we found the capsule size, amount of PCM, pump flow and arrangement that give a 50 L solar tank the most extra hot water at 50 C between 18:00 and 09:00, compared with the same tank holding water only. A physics simulator scores every design over full years of real weather; DOE, surrogate models and NSGA-II decide which designs to simulate; a pre-declared rule picks the design on separate years; a one-time test on untouched years (2024-2025) confirms it. **PCM helps in regimes 0-2 (56-76 kWh/yr more on the test years); in regime 3 (cold, high altitude) it does not, so a water-only tank is recommended there.**

## 1. The problem

- **Inputs from Objective 1:** 4 regimes (medoids UKP_0014 / 0029 / 0025 / 0041; mains water 21.8 / 18.6 / 17.0 / 7.4 C) and 3 shortlisted PCMs each, so **12 (regime, PCM) pairs**.
- **Fixed system:** 50 L tank, 1.5 m2 flat-plate collector, 0.8 mm aluminium capsules, 300 L/day demand in local time, delivery at 50 C, limits water 75 C and PCM 65 C.
- **Designed:** capsule inner radius 10-40 mm, PCM fraction 2-30 %, flow 0.010-0.050 kg/s, arrangement (single-layer, staggered, radial), and the PCM.
- **Weather years:** explore 2016-2020 (search), select 2021-2023 (choose), test 2024-2025 (used once).

## 2. What the simulation includes

| Part | Model |
|---|---|
| Collector | Hottel-Whillier-Bliss equation; pump off below 10 W/m2 |
| Tank | 10 stacked layers (stratified): hot draw from the top, mains in at the bottom, collector return to the layer of equal temperature, wall losses, mixing of inverted layers |
| PCM capsules | Enthalpy method (melting band Tm +/- 1 K); PCM mass from the inner sphere; wall heat capacity included |
| Water to PCM | Wakao-Kaguei water-side coefficient + aluminium wall + R_pcm = r/(5k), x2 once half melted (convection) |
| Demand, mains, shield | Hourly profile in local time (peak 07:30); mains per regime; charging bypass at 72 C water / 62 C PCM |
| Solver and checks | 150 s linear-implicit step (numba-compiled); energy residual below 0.1 %; time-step error 0.029 %; Phase-4 gates passed |

**Main metric:** g_window = energy delivered at 50 C from 18:00 to 09:00 by the PCM tank minus the water-only tank at the same flow and weather year. Also g_annual and d_overheat (extra shield hours). **Not modelled:** supercooling, volume change, backup heater, pump energy.

## 3. Algorithms used and why

| Step | Algorithm used | Why | Not used, and why |
|---|---|---|---|
| Simulator | Grey-box physics model, about 1 s per year | Fast enough for tens of thousands of simulated years; every equation is standard and traceable | CFD (weeks per year of weather), TRNSYS (not callable thousands of times) |
| PCM | Enthalpy method | Conserves latent heat exactly; never fails at the melting point | Apparent heat capacity (loses latent heat at 150 s), front tracking (too slow) |
| Tank | 10-layer stratified tank | Real tanks keep hot water on top: 22.5 L delivered vs 27 L in CFD; a mixed tank gives 0 | Single mixed node; 20+ layers (under 1 % change, double cost) |
| Time step | Linear-implicit (backward Euler), 150 s | Stable without iterations; verified to 0.029 % | Explicit (about 50x slower), Runge-Kutta (tiny steps at melting) |

| Step | Algorithm used | Why | Not used, and why |
|---|---|---|---|
| Comparison | Paired same-flow water-only tank, PCHIP interpolation | Isolates the effect of the PCM from flow and weather | Absolute energy (flow would look like a PCM effect) |
| Years | Explore / select / one-shot test | No selection bias; an honest final number | Using all years; random k-fold |
| Screening | Taguchi L9 + ANOVA | 9 runs instead of 81; standard in PCM-tank papers | Full factorial, Plackett-Burman (no curvature) |
| DOE | Scrambled Sobol, 128 points | Even coverage of the design box with few runs | Random sampling, Latin hypercube |
| Surrogate | XGBoost / quadratic RSM / NN ensemble, chosen by rank validation (XGBoost 11 pairs, RSM 1; same as Rajasthan) | Chosen by data; only proposes designs, never produces a reported number | Kriging (fair alternative, not in the recipe), RBF, SVR, deep networks (too little data) |
| Real gain | Measured repeatability band (about 22 / 19 / 20 / 6 kWh) | Includes weather, time step, tank layers and property uncertainty | p-values (over-confident), fixed % |
| Optimiser | NSGA-II (100 x 100, 3 seeds) + active learning + box-corner step | Three conflicting objectives without weights; corrects the surrogate where unsure | Weighted sum, Bayesian optimisation, PSO |
| Check | Dense grid of 720 designs per regime, hypervolume and epsilon | Proves no better region was missed (all 4 regimes within the band) | Trusting the optimiser unchecked |
| Choice | Band-based rule, then lowest PCM mass | Written before running; no weights; can honestly say 'no gain' | Best-only, TOPSIS, weighted score |
| Final test | One-shot held-out test, locked | A test that cannot be repeated or tuned | Cross-validation only |
| Controls | Ablations and controls | Latent heat set to zero removes 139-155 kWh of gain: the gain is the PCM | - |
| CFD check | OpenFOAM enthalpy-porosity, axisymmetric wedge | Free, scriptable; checks the capsule model | Fluent, COMSOL (commercial, same physics) |

## 4. Running it faster (parallel tasks and CPU)

Measured on this laptop: i7-13700H, 14 cores / 20 threads, 15.7 GB RAM. One simulated year 1.09 s; about 180 MB per process; whole chain about 6 h with 6 parallel jobs.

| Method | Effect | Status |
|---|---|---|
| 12 (regime, PCM) pairs as separate processes (pipeline.py --jobs) | Up to N times faster for the pair steps | Used (6 jobs); 10-12 possible if memory allows |
| 5 weather years of a design in parallel | Measured 3.8x (5.43 s to 1.42 s, same result) | Proposed, not in code |
| Read the hour of the day as one array, not row by row | About 25 % faster per year (same values) | Proposed, not in code |
| numba-compiled time-step loop; cached weather and baselines | Already fast | Used |
| One numpy thread per process (OMP_NUM_THREADS=1); charger in, Best performance mode | Avoids slow-downs | Practical |
| OpenFOAM cases 4-5 at a time | About 4-5x for 26 cases | Ready to run now |

**Estimate:** combined, the 6 h chain could drop to about 2-3 h (not measured end to end). **Does not help:** GPU (each time step depends on the previous one), more than 14 processes, MPI for small OpenFOAM cases, a larger time step.

> **Do not re-run the steps that choose the design.** The results are final and the 2024-2025 test has been used (results/held_out_test.lock). Only the reporting stages and the OpenFOAM cases are safe to run now.

## 5. Results and final output

| Regime | Chosen design | Select years | Test 2024-25 | Verdict |
|---|---|---|---|---|
| 0 | PureTemp 58, R 39.8 mm, 26.3 % PCM, 47 capsules, 0.019 kg/s, staggered | 80.0 kWh/yr | 76.0 kWh/yr | PCM helps |
| 1 | PureTemp 58, R 39.3 mm, 29.7 %, 55 capsules, 0.031 kg/s, single-layer | 72.1 | 61.8 | PCM helps |
| 2 | savE OM55, R 39.7 mm, 30.0 %, 54 capsules, 0.020 kg/s, single-layer | 63.0 | 56.4 | PCM helps |
| 3 | (least-PCM design: n-Hexacosane, 2 %, 4 capsules) | -5.8 | -6.1 | Water-only tank |

- All designs safe: PCM at most 64.2 C (limit 65), water at most 73.1 C (limit 75). The best designs use large capsules (about 40 mm) and close to the 30 % PCM limit.
- **CFD check (1 of 10 capsule cases run):** regime 0 capsule 90 % melted in 71 min in CFD vs 145 min in the simulator, so the simulator is conservative.
- **Deliverables:** the chosen designs (results/g10_selection_r*.json, held-out results), the Objective 3 hand-off contract (results/obj3_environment_contract_uttarakhand_v2.json), the main report Objective2_Uttarakhand_Report_v3, the full algorithms document, and the OpenFOAM folder.